COMPUTING THE DERIVED GLOBAL SECTIONS AND DERIVED HOM AGAINST A PERFECT OBJECT ALONG THE DESCENT TOWER

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a stage-wise RHom computation with base change compatible perfect transitions, the derived-limit commutation of RHom_X(E, -), and a duality translation from the hom statement to the tensor statement

Editorial note. The label lemma-compute-against-perfect names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The descent tower situation with its system of objects and the object K defined as the derived limit of the system, and the identification of RHom with RGamma composed with the internal Hom sheaf
- The base change lemma for RHom against a perfect object: the stages RHom_{X_n}(E_n, F_n) are perfect over A_n with formation commuting with base change
- The More on Algebra lemma that a derived limit of perfect objects with compatible base change data is perfect and recovers the stages on base change
- The duality between derived global sections and derived hom translating tensor statements into hom statements for perfect objects

Source reference. Original reference 0CTD: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 2388–2388. Prerequisite pointer invoked at proof line(s) 2407; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0CQG: sources/stacks-chow/tags.txt, source lines 15961–15961. Prerequisite pointer invoked at proof line(s) 2436; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08JJ: sources/stacks-chow/tags.txt, source lines 10852–10852. Prerequisite pointer invoked at proof line(s) 2451; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-flat.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original section 0B6E: sources/stacks-chow/tags.txt, source lines 14067–14067. Prerequisite pointer invoked at proof line(s) 2429; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08B7: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 2362–2362. Prerequisite pointer invoked at proof line(s) 2406; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0A1R: sources/stacks-chow/tags.txt, source lines 12679–12679. Prerequisite pointer invoked at proof line(s) 2431; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 08B7 let K be as in Lemma 0CTD. For any perfect object E of $D(\mathcal{O}_X)$ we have*

1. *$M = R\Gamma(X, K \otimes^{\mathbf{L}} E)$ is a perfect object of $D(A)$ and there is a canonical isomorphism $R\Gamma(X_n, \mathcal{F}_n \otimes^{\mathbf{L}} E|_{X_n}) = M \otimes^{\mathbf{L}}_A A_n$ in $D(A_n)$,*
2. *$N = R\mathrm{Hom}_X(E, K)$ is a perfect object of $D(A)$ and there is a canonical isomorphism $R\mathrm{Hom}_{X_n}(E|_{X_n}, \mathcal{F}_n) = N \otimes^{\mathbf{L}}_A A_n$ in $D(A_n)$.*

In both statements $E|_{X_n}$ denotes the derived pullback of E to X_n.

Proof. Proof of (2). Write $E_n = E|_{X_n}$ and $N_n = R\mathrm{Hom}_{X_n}(E_n, \mathcal{F}_n)$. Recall that $R\mathrm{Hom}_{X_n}(-,-)$ is equal to $R\Gamma(X_n, R\mathcal{H}om(-,-))$, see Cohomology on Sites, Section 0B6E. Hence by Derived Categories of Spaces, Lemma 0A1R we see that N_n is a perfect object of $D(A_n)$ whose formation commutes with base change. Thus the maps $N_n \otimes^{\mathbf{L}}_{A_n} A_{n-1} \to N_{n-1}$ coming from φ_n are isomorphisms. By More on Algebra, Lemma 0CQG we find that $R\lim N_n$ is perfect and that its base change back to A_n recovers N_n. On the other hand, the exact functor $R\mathrm{Hom}_X(E, -) : D_{QCoh}(\mathcal{O}_X) \to D(A)$ of triangulated categories commutes with products and hence with derived limits, whence

$$R\mathrm{Hom}_X(E, K) = R\lim R\mathrm{Hom}_X(E, \mathcal{F}_n) = R\lim R\mathrm{Hom}_X(E_n, \mathcal{F}_n) = R\lim N_n$$

This proves (2). To see that (1) holds, translate it into (2) using Cohomology on Sites, Lemma 08JJ. □